

Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

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Triples: 10 · Verdicts: 10 · Inconsistencies: 0 ·
Violations: 0

EXECUTIVE SUMMARY

1 Correct	6 Rule Violation	1 Unverifiable
1 Predicate Mismatch	1 Fabricated Fact	

EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response before canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
300	HAS_INTEREST_RATE	20	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with a particular entity.
850	HAS_INTEREST_RATE	20	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with a particular entity.
333	HAS_INTEREST_RATE	30	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with a particular entity.
300	HAS_RISK_TIER	near prime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to a particular entity.
850	HAS_RISK_TIER	prime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to a particular entity.
305	HAS_RISK_TIER	subprime	Yes	FACTUAL_CLAIM	Indicates the risk tier classification assigned to a particular entity.
paris	IS_THE_CAPITAL_OF	france	Yes	FACTUAL_CLAIM	Indicates the country of which a city is the capital.
850	HAS_DOWN_PAYMENT	4	Yes	FACTUAL_CLAIM	Indicates the down payment amount associated with a particular entity.
850	HAS_DOWN_PAYMENT	2	Yes	FACTUAL_CLAIM	Indicates the down payment amount associated with a particular entity.
335	HAS_DOWN_PAYMENT	2	Yes	FACTUAL_CLAIM	Indicates the down payment amount associated with a particular entity.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. Shows original predicate, canonical match, and match confidence.

Subject	Original Predicate	Object	Canonical Predicate	Confidence
300	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
850	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
333	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
300	HAS_RISK_TIER	near prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
850	HAS_RISK_TIER	prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Confidence
305	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
paris	IS_THE_CAPITAL_OF	france	IS_THE_CAPITAL_OF	LOW
850	HAS_DOWN_PAYMENT	4	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
850	HAS_DOWN_PAYMENT	2	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
335	HAS_DOWN_PAYMENT	2	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of the database.

No logical inconsistencies detected.

SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

No symbolic rule violations detected.

DETECTED HALLUCINATIONS

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	Rule Violation	STRICT	OBJECT	1.00	Subject and predicate match DB, but object '20' contradicts stored value '22.94'. Evidence: [{"subject": "300", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "22.94"}]
850	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	Rule Violation	STRICT	OBJECT	1.00	Subject and predicate match DB, but object '20' contradicts stored value '4.36'. Evidence: [{"subject": "850", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "4.36"}]
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	Fabricated Fact	NONE	SUBJECT	0.20	Predicate exists in DB schema, but neither subject '333' nor object '30' appear in any triple with this predicate. Evidence: []
300	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	near prime	Rule Violation	SEMANTIC	OBJECT	1.00	Subject and predicate match DB, but object 'near prime' contradicts stored value 'deep subprime' (different risk tiers). Evidence: [{"subject": "300", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime"}]
850	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	prime	Rule Violation	SEMANTIC	OBJECT	1.00	Subject and predicate match DB, but object 'prime' contradicts stored value 'super prime' (different risk tiers). Evidence: [{"subject": "850", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "super prime"}]
305	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	subprime	Rule Violation	SEMANTIC	OBJECT	1.00	Subject and predicate match DB, but object 'subprime' contradicts stored value 'deep subprime' (different risk tiers). Evidence: [{"subject": "305", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime"}]

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
paris	IS_THE_CAPITAL_OF	france	Unverifiable	None	—	0.00	No DB evidence found and predicate is unknown in schema; cannot confirm or deny the triple. Evidence: []
850	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	4	Rule Violation	Strict	OBJECT	1.00	Subject and predicate match DB, but object '4' contradicts stored value '2.0'. Evidence: [{"subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2.0"}]
850	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	Correct	Semantic	—	1.00	All components match DB; object '2' semantically equals stored value '2.0' (same numerical value). Evidence: [{"subject": "850", "predicate": "DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION", "object": "2.0"}]
335	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	Predicate Mismatch	Strict	PREDICATE	0.60	Subject '335' exists in DB, but predicate 'DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION' is not a valid relation for it. Evidence: [{"subject": "335", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime"}, {"subject": "335", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "20.88"}]

TRIPLE AUDIT TRAIL — ORIGINAL → CANONICAL → VERDICT

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict — e.g. dropped due to output truncation.

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
300	HAS_INTEREST_RATE	20	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	RULE_VIOLATION
850	HAS_INTEREST_RATE	20	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	RULE_VIOLATION
333	HAS_INTEREST_RATE	30	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	FABRICATED_FACT
300	HAS_RISK_TIER	near prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DEFAULT_LEVEL	near prime	RULE_VIOLATION
850	HAS_RISK_TIER	prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DEFAULT_LEVEL	prime	RULE_VIOLATION
305	HAS_RISK_TIER	subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DEFAULT_LEVEL	subprime	RULE_VIOLATION
paris	IS_THE_CAPITAL_OF	france	LOW	IS_THE_CAPITAL_OF	france	UNVERIFIABLE
850	HAS_DOWN_PAYMENT	4	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	4	RULE_VIOLATION
850	HAS_DOWN_PAYMENT	2	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	CORRECT

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
335	HAS_DOWN_PAYMENT	2	HIGH	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2	PREDICATE_MISMATCH

VERDICT LEGEND

Verdict	Description
Correct	All three components (subject, predicate, object) verified against DB.
Rule Violation	One or more components contradict DB evidence.
Multi-Rule Violation	Violation spans multiple DB triples simultaneously.
Unverifiable	No usable DB evidence found for any component.
Predicate Mismatch	Subject real, but stated predicate is not a valid relation for it.
Fabricated Fact	Triple asserts a relationship that does not exist anywhere in DB.